Non-Anticipative Market Field Calculus: An Auditable Multiscale Representation for Option-Path Decision Support

Anonymous authors
Paper under double-blind review

Abstract

We introduce Market Field, a non-anticipative multiscale representation of market path organization designed for interpretable research and prospective option-decision support. OHLCV bars are transformed into a bounded horizon–time pressure surface; paper experiments and option snapshots use completed bars, while the live research screen may include a provider's forming bar. Prefix-only temporal differences, derivatives over log horizon, ordinal disorder, propagation analogues, and OHLCV carriers produce a compact state vector. A deterministic, request-local Form dictionary translates that vector into calibration-relative descriptions while preserving a strict fit–calibration–evaluation chronology. A production integration attaches completed-daily-bar snapshots to option scanner events and position reviews, but assigns them zero algorithmic ranking weight and no execution authority; displayed advice may still influence a human. Across 32 tested daily prefix endpoints, comprising 6,688 serialized channel–horizon scalar checks, we observe no mismatches at 10^{-4} output precision, and all eight fixed-input dictionaries reproduce exactly. A broader nine-timeframe audit passes 46/46 full-precision prefix checks covering 24,472 live numeric values with maximum error zero at tolerance 10^{-12}. Controlled paths produce ordered horizon-response delays ($\rho = 1.000$). Under the chosen weighted distance, support rule, and silhouette gate, multi-Form codebooks are accepted for two of eight assets. State-conditional upper-distance-tail rates vary materially across assets and windows. These tests exercise selected software invariants and representation behavior; they do not establish forecast skill, option profitability, or a physical law of markets. We provide source, hashes of locally retained inputs, derived results, figures, tables, and executed notebooks, while provider restrictions prevent redistribution of the raw snapshots.

1 Introduction

Market analysis interfaces usually compress a path into a small set of single-window indicators. This makes the output easy to scan but obscures whether a move is coherent across horizons, confined to one scale, accelerating, fragmenting, or changing its relationship to activity and liquidity. The opposite response—adding more indicators—often produces a wall of numbers with no stable visual or semantic grammar. Both patterns become especially problematic for options: a volatility-dislocation thesis and the underlying price path are distinct objects, yet they are often blended into one opaque score.

We study a narrower question: can a multiscale state representation remain non-anticipative, inspectable, visually legible, and safe to introduce into an existing decision system before economic efficacy is known? Our answer is Market Field, an engineered field over time t and lookback horizon h. The word field describes the data structure, not a claim that prices obey fluid mechanics. Likewise, pressure, velocity, acceleration, jerk, and snap are names for bounded discrete measurements, not literal forces or hidden physical dimensions.

The system evolved from SwamiChart-style horizon strips (Ehlers & Way, 2012a;b) through a signed pressure surface, a derivative and scale geometry, a human translation layer, and finally a versioned shadow integration for option research. This progression preserves the original multiscale visual intuition while replacing indicator voting with an explicit measurement pipeline.

Version boundary. market_field_calculus_v1 names the formulas evaluated here. Additive semantic_revision=1.1 names aliases, maturity/input-quality metadata, alignment precedence, authority, and effect receipts that leave that formula vector unchanged. market_field_preliminary_v2 names the expanded evaluation harness, not a v2 field model; formula-changing alternatives remain explicit future ablations.

Our contributions are:

1. a non-anticipative bounded pressure surface over causal time and log-horizon coordinates, with directional, kinematic, geometric, information, propagation, and OHLCV carrier strata;
2. a deterministic empirical Form dictionary whose robust scaling, model selection, calibration evidence, and evaluation sequence are chronologically separated;
3. a human-facing visual grammar combining a narrow horizon cloud with engineered phase portraits, while explicitly distinguishing trajectories from cycles or reconstructed attractors;
4. a prospective options interface that stores point-in-time field snapshots with immutable decision history under zero-rank, no-execution safeguards; and
5. a reproducible diagnostic suite that tests prefix invariance, determinism, controlled representation behavior, codebook restraint, and local compute cost without converting those checks into a performance claim.

2 Relationship to Prior Work

Multiscale representation. SwamiCharts arrange an indicator across many lookbacks (Ehlers & Way, 2012a;b). Wavelets, scattering transforms, and multifractal models offer principled multiscale representations (Mallat, 2012; Andén & Mallat, 2014; Bacry et al., 2001). Market Field shares their multiscale motivation but does not implement a wavelet transform, scattering network, or multifractal estimator. Its horizon coordinate is a set of EMA/true-range/path-efficiency measurements selected for causal interactive computation.

State and phase descriptions. Markov switching, structural-break, and online change-point models estimate latent or discontinuous states (Hamilton, 1989; Bai & Perron, 1998; Adams & MacKay, 2007). Market Field instead creates a calibration-relative codebook with deterministic clustering. Its scope plots are engineered two-dimensional projections. They are visually related to phase-space and recurrence analysis (Takens, 1981; Eckmann et al., 1987; Marwan et al., 2007), but they are not delay-coordinate reconstructions, recurrence plots, attractor tests, or cycle detectors. Permutation entropy is the one directly adopted nonlinear statistic (Bandt & Pompe, 2002).

Technical and option context. Prior work gives statistical definitions to technical patterns and price barriers (Lo et al., 2000; Chung & Bellotti, 2021). The present support and resistance layer is deliberately simpler: shifted 20-bar extrema used as causal context, not learned order-book structure. Implied-versus-realized volatility and variance-risk-premium measures are established (Carr & Wu, 2009; Bollerslev et al., 2009; Goyal & Saretto, 2009). Our open question is whether separately measured path fit conditions such optionality signals, not whether the volatility spread itself is novel. Similarly, cross-market dependence and connectedness are well studied (Diebold & Yilmaz, 2012; Baruník & Křehlík, 2018); the current cached-source atlas is an association screen, not causal discovery.

Validation discipline. Financial feature search is exposed to dependence, selection, and multiple testing (White, 2000; Harvey et al., 2016; Bailey et al., 2017). We therefore separate representation diagnostics from economic evaluation and treat rolling-origin, purged, cost-aware option tests as future work. This distinction is central: a correctly non-anticipative implementation can still have no predictive or economic value.

3 Non-Anticipative Market Field

3.1 Pressure over time and horizon

Let O_t, H_t, L_t, C_t, V_t denote a sorted, deduplicated OHLCV prefix, let $M_t = (H_t + L_t)/2$, and index an ordered horizon grid by $h_1 < \cdots < h_J$. For row j, a gap-aware true range normalizes the displacement between a fast EMA with span $\max(2, \lfloor h_j/2 \rfloor)$ and an EMA with span h_j:

$$z_{j,t} = \frac{\mathrm{EMA}_{\max(2,\lfloor h_j/2\rfloor)}(M)_t - \mathrm{EMA}_{h_j}(M)_t}{\mathrm{Mean}_{h_j}(TR)_t}, \qquad d_{j,t} = \frac{z_{j,t}}{1 + |z_{j,t}|}, \tag{1}$$

$$e_{j,t} = \mathrm{clip}_{[0,1]} \frac{|M_t - M_{t-h_j}|}{\sum_{i=t-h_j+1}^{t} |M_i - M_{i-1}|}. \tag{2}$$

Equation 2 is Kaufman's efficiency-ratio construction (Kaufman, 1995). The raw signed state $d_{j,t}e_{j,t}$ is smoothed causally in time and blended only with adjacent rows to obtain $P_{j,t}$. Thus P is bounded, directional, and attenuated when net displacement is small relative to path length. Row adjacency is part of v1: it is not a log-distance kernel, so changing horizon density changes the effective model as well as its discretization. The blend is analytical, and all downstream measurements are calculated from the blended surface. Exact coefficients, initialization, zero conventions, and edge behavior appear in Appendix A.

Proposition 1 (boundedness and prefix measurability). Let $\mathcal{H} = (h_1, \ldots, h_J)$ be fixed. Assume finite valid OHLC inputs, positive closes, the numerical conventions in Appendix A, and EWM recursions initialized from the observed prefix only. Then $e_{j,t} \in [0, 1]$, $d_{j,t}, P_{j,t}, D_q(X)_{j,t} \in (-1, 1)$ except for their defined zero cases, clipped unsigned channels lie in $[0, 1]$, and every live field output at t is a deterministic function of the OHLCV prefix $X_{1:t}$.

Proof. The triangle inequality bounds endpoint displacement by total path length. The map $x/(1 + |x|)$ is bounded; prefix-initialized EWMs and the stated nonnegative blends are convex; and past-only rolling windows and recursions preserve prefix measurability by induction in t. Clipping preserves the asserted unsigned bounds. The zero-denominator conventions close the degenerate cases. This result covers the live field and its point-in-time option snapshot, not the window-refit Form dictionary, retrospective outcome atlas, cross-market cache, or any future label. The 32 endpoint audits in Section 5 are regression tests of the implementation of this property, not a statistical proof of causality.

3.2 Time/scale calculus and strata

For any series X, define a causal bounded difference

$$D_q(X)_t = \frac{r_t}{1 + |r_t|}, \qquad r_t = \frac{\Delta X_t}{\mathrm{EWM}_q(|\Delta X|)_t}. \tag{3}$$

Successive applications form velocity $v = D_{13}(P)$, acceleration $a = D_{13}(v)$, jerk $j = D_{13}(a)$, and snap $s_4 = D_{13}(j)$. Numerical derivatives across $u_j = \log h_j$ yield scale gradient g, scale curvature c, and the bounded temporal change of scale gradient m. These are normalized finite differences; the kinematic vocabulary is mnemonic. Appendix A specifies the nonuniform-grid interior and edge stencils and every horizon reduction used to obtain the state vector.

The derivative surfaces are reduced into five interpretable strata. Structure combines signed-state strength and cross-horizon coherence. Kinematics weights $|v|, |a|, |j|, |s_4|$. Geometry

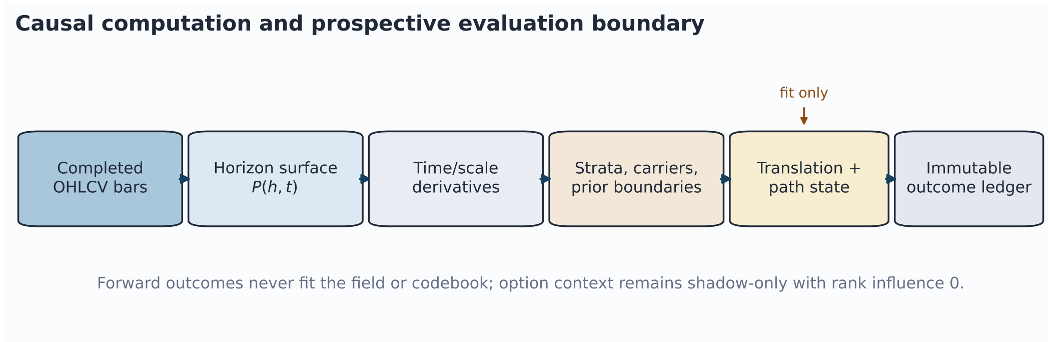

Figure 1: The auditable computation boundary. Audited experiments and option snapshots use completed bars; outcomes are recorded later and never refit the current field. The live research screen may include a provider's forming bar. The option integration has zero algorithmic rank and execution authority, although a displayed advisory may influence a human.

combines $|g|$, $|c|$, $|m|$, and boundary energy. Information combines causal order-3 Bandt–Pompe entropy with an engineered disorder channel. Propagation summarizes co-occurring time and scale change; its signed cascade bias indicates the orientation of that change. Separate carriers report realized variation, volume participation, and an Amihud-like price-impact ratio (Amihud, 2002). All are descriptive measurements. In particular, organization is not a probability, propagation is not causal transmission, and the scaling exponent is not a Hurst exponent.

3.3 A calibration-relative language

The research interface aggregates five derivatives, seven strata, and three carriers into $\mathbf{x}_t \in \mathbb{R}^{15}$. Derivative, stratum, and carrier families each receive one third of total squared-distance weight. Robust center and scale are estimated only on the proper-fit segment. A deterministic farthest-point-initialized k-means procedure (MacQueen, 1967; Gonzalez, 1985) considers at most five Forms, requires each cluster to contain at least $\max(20, \lceil 0.05n \rceil)$ fit bars, and accepts a multi-Form codebook only when mean silhouette is at least 0.25 (Rousseeuw, 1987). Otherwise it emits one Form. The initialization is a deterministic variant, not MacQueen's original seeding rule; exact candidate selection and tie handling are given in Appendix B.

The later calibration segment provides state-conditional nearest-centroid distances. For evaluation distance d_t in state k,

$$\tau_t = \frac{1 + \#\{d_i^{\mathrm{cal},k} \geq d_t\}}{n_k + 1}. \tag{4}$$

The v1 payload's legacy key outside_learned_range is true when $\tau_t < 0.05$ and $n_k \geq 20$; throughout this paper we call that event the upper state-conditional calibration-distance tail. It does not imply coordinatewise range violation, convex-hull exclusion, density support, or a forecast. Equation 4 is a discrete empirical rank diagnostic: for $n_k = 20$ its minimum is $1/21$, so the strict 0.05 rule fires only beyond every same-state calibration distance. Form identifiers are request-local: F.001 for one symbol/window has no asserted identity elsewhere.

The visual translation layer shows the horizon cloud and three scopes: direction (P, v) colored by structure, structure–information colored by kinematics, and propagation–cascade bias colored by geometry. The second scope does not plot the legacy organization channel O. Display-only smoothing can soften the trace, but the raw path remains available. Loops are trajectories, not detected cycles.

4 Prospective Option Integration

The key design separates option mispricing from underlying path fit. Existing scanner eligibility and the champion opportunity score remain functions of cheapness, volatility edge, contract quality, execution quality, and a pre-existing recurrence term. A field snapshot is constructed only after an option already qualifies under the existing optionality logic. It cannot create eligibility or alter algorithmic rank.

The original v1 contract interprets scanner candidates as long, directional, single-leg calls or puts: calls take side $r = +1$ and puts $r = -1$. The additive v1.1 semantic contract instead uses, in order, signed net delta, explicit opening action plus option type, or the legacy long-single-leg assumption; unsupported exposure abstains. Aggregate pressure and velocity are aligned as $p^\star = rp$ and $v^\star = rv$. The advisory state is mixed when $|p^\star| \leq 0.02$, contradictory when $p^\star < 0$, fading when $p^\star > 0$ and $v^\star < 0$, and supportive otherwise. Separately, the kinematic-exhaustion hypothesis uses pressure-aligned velocity $\widetilde{v} = \mathrm{sign}(p)v$. Four exploratory hypotheses—organized expansion, longward cascade, geometry/disorder shock, and kinematic exhaustion—use quantiles computed from prior rows only. The precise conditions appear in Appendix C.

Every snapshot states its contract: schema option_market_field_v1, model market_field_calculus_v1, additive semantic_revision=1.1, legacy mode label shadow_only, rank influence 0, and automatic execution false. A machine-readable authority object assigns no scanner-rank, hard-veto, manager-verdict, target-size, or execution authority while marking displayed confidence and review priority as advisory. It excludes the retrospective Form atlas, future outcomes, cross-market caches, and option-derived inputs. "Shadow" therefore means zero algorithmic scoring and execution authority, not zero behavioral influence: a human may act on the display. Scanner events, position entry, reassessments, and their field snapshots are append-only. Any prospective outcome study must record whether the user saw the field and what action followed, or hide the field during collection; otherwise treatment exposure can contaminate discretionary closure outcomes. At present, cohorts are descriptive and do not control algorithmic decisions.

5 Diagnostic Evaluation

5.1 Questions, data, and protocol

We evaluate representation mechanics, not trading performance:

RQ1 Is the serialized field prefix-invariant and the learned dictionary deterministic?

RQ2 Do controlled paths produce the intended directional, organizational, and cross-horizon behavior?

RQ3 Does the codebook decline weak separation, and how does its range diagnostic behave across assets?

RQ4 What local cache-only compute and payload scale does a compact option snapshot require?

The primary audit uses adjusted daily Yahoo Finance OHLCV from 2018-01-01 through 2026-07-21 for SPY, QQQ, IWM, TLT, GLD, USO, VNQ, and BTC-USD. Each normalized local snapshot is hashed; the manifest records requested bounds, actual bounds, row count, adjustment mode, and SHA-256. The first-run manifest records cd67fb4 as its repository reference together with source-file hashes; it must not be read as the identity of later reruns. The current primary receipt records clean source HEAD b1755d2 plus source and artifact hashes. The current supplementary receipt records a dirty working tree and parent HEAD f0ef66a; its per-file start/completion hashes, not that parent commit, identify the evaluated source, and it reports whether any source changed during the run. The companion script can rerun exactly only when the local raw cache is retained, or retrieve a fresh provider snapshot whose hashes and results may differ. Provider terms prevent redistribution of the raw CSVs; the public package contains the manifest, hashes, derived results, and executed notebooks.

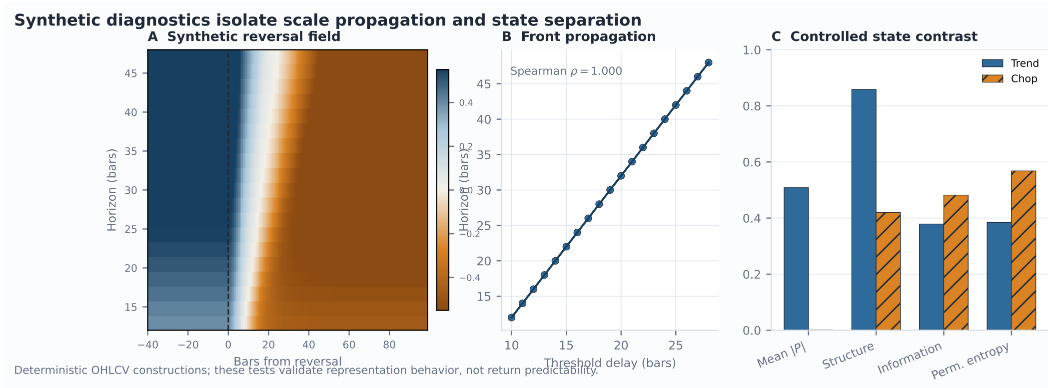

Figure 2: Controlled diagnostics. (A) A deterministic reversal moves through the horizon surface. (B) the threshold-crossing delay increases monotonically with horizon. (C) the intended trend/chop contrast appears in pressure magnitude, structure, information, and permutation entropy. These are construction tests, not return forecasts.

Diagnostic	Result	Interpretation
Prefix invariance	6,688 comparisons, 0 mismatches	No future-bar dependence detected at 10^{-4} API precision.
Deterministic dictionary	8/8 exact reruns	Fixed inputs yield byte-identical versioned lexicons.
Synthetic propagation	$\rho = 1.000$	Longer horizons cross the reversal threshold later by construction.
Distance-tail diagnostic	1.5–10.6% by asset	State-conditional empirical ranks are heterogeneous and non-exchangeable.
Codebook support gate	2/8 multi-Form	Multi-Form solutions appear only when the declared metric, support, and silhouette gates pass.
Compact snapshot	106.5 ms median; 3,081 bytes	Local compute-only benchmark; ranking and execution remain disabled.

Table 1: Representation and operational diagnostics. Exact-zero results apply at serialized precision; timing is a local compute-only benchmark.

Thus a hash identifies the original input but does not make that input recoverable from a fresh clone. A supplementary locally retained audit covers 15 datasets and all nine supported SPY timeframes, tests horizon-grid steps 1, 2, 4, and 8, and probes dictionary window stability (Appendix E).

For prefix invariance, we compare 11 channels across 19 horizons at four historical cut points per asset against the same timestamps computed with the complete suffix present. These are 32 prefix endpoints with 209 related, four-decimal serialized values at each endpoint, not 6,688 independent tests. Determinism reruns the full versioned lexicon and compares canonical JSON hashes. A deterministic 800-bar construction reverses log drift at bar 400; a second construction contrasts an organized trend with alternating chop. We also time 80 completed-daily option snapshots over eight symbols using 365 in-memory bars, explicitly excluding retrieval, persistence, cold start, and concurrency.

5.2 Results

Across all 32 primary endpoints, the 6,688 serialized scalar comparisons are identical at 10^{-4} precision, with maximum observed serialized difference zero. The supplementary audit bypasses response rounding and passes 46/46 prefix checks spanning 24,472 full-precision values from live field matrices and aggregate derivatives, strata, carriers, and carrier ratios; maximum absolute error is zero at tolerance 10^{-12}. It does not yet exhaust dictionary

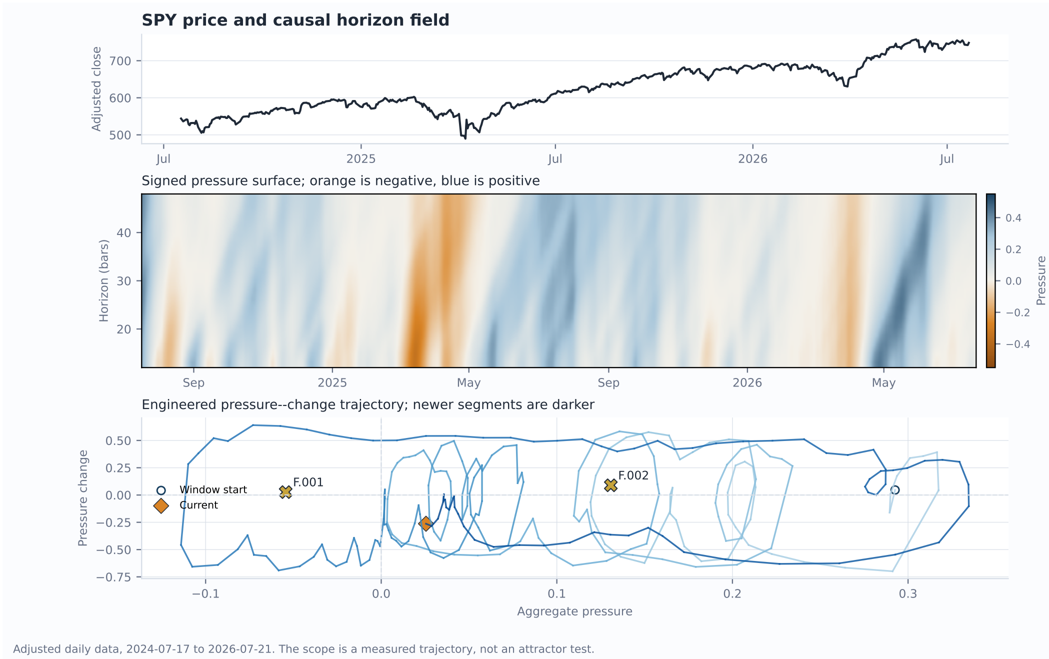

Figure 3: SPY example over the final 500 daily bars. Top: adjusted close. Middle: signed horizon field. Bottom: a pressure–change trajectory and request-local Form centroids. The phase portrait is an engineered projection, not an attractor test.

assignments, maturity metadata, hypothesis flags, or every downstream option field. All eight tested primary lexicons reproduce byte-for-byte. On the synthetic reversal, threshold-crossing delays range from 10 to 28 bars over horizons 12 through 48, with Spearman $\rho = 1.000$. The organized segment has mean $|P| = 0.508$ and structure 0.859, versus 0.001 and 0.419 for alternating chop; permutation entropy increases from 0.385 to 0.568. The nonzero trend entropy reflects entropy of transformed pressure rows including startup and transition behavior, not entropy of a strictly monotone price sequence.

Under the chosen weighted Euclidean representation, k-means family, minimum-support rule, and silhouette threshold, multi-Form codebooks are accepted only for SPY (two) and USO (three). This may reflect weak separation, nonspherical structure, temporal drift, limited support, or a conservative gate; it is not proof that other assets have one natural state. Among 7,237 evaluation bars with state-conditional support, 395 enter the upper 5% calibration-distance tail (5.46% pooled). Rates vary from 1.51% for VNQ to 10.58% for SPY (Figure 4); 33 USO evaluation bars lack the required same-state support and are reported separately from the denominator. Serial dependence, distribution shift, discrete ranks, and state conditioning prevent a nominal coverage interpretation. The shorter-window rates and Form instability in Appendix E are more consequential than the pooled proximity to 5%.

Resolution sensitivity is material. Relative to a dense step-1 horizon grid over 8–64 bars, the supplementary step-4 audit has median correlation 0.940 but median IQR-normalized MAE 0.203 across 180 dataset–feature pairs; the 95th percentile error is 0.428. The reported total of 540 covers all three coarser grids (steps 2, 4, and 8), not step 4 alone. Pressure is comparatively stable, while geometric and propagation summaries move more. Because row-adjacent blending also changes with the grid, horizon density is a model parameter, not a rendering preference or pure numerical-resolution choice.

The run-specific primary compute and payload measurements appear in Table 1; the current supplementary distribution is preserved row-by-row in option_snapshot_latency.csv and linked to its source hashes by the run receipt. These volatile local measurements show feasibility for cache-only instrumentation, not a stable latency or payload contract. They exclude retrieval, persistence, fresh-process import, concurrent load, network and queueing, and a declared budget; live endpoint timing is a separate operational artifact. The emitted payloads retain rank influence 0 and automatic execution false.

6 Discussion and Limitations

The diagnostics support a modest claim: the serialized implementation is prefix-invariant at the tested endpoints, deterministic for fixed inputs, responsive to controlled path structure, and computationally plausible in a cache-only local benchmark. They do not show that the representation predicts returns, improves option outcomes, or meets a production latency service level.

Several limitations are material. Parameters, horizon sets, smoothing coefficients, aggregation weights, clustering gates, and hypothesis quantiles are hand designed. High-order differences amplify noise, and the grid audit shows that some derived channels are resolution-sensitive. Row-adjacent smoothing is not invariant to log-horizon spacing. The research page may include a provider's partial bar, although the option snapshot removes incomplete daily bars using a fixed 16:15 America/New_York cutoff that is not exchange-calendar-aware. The public API requests a hidden prefix of $\max(72, 2h_{\max})$ bars beyond the visible response, but providers may return fewer bars. V1.1 reports visible versus calculation bars, input-quality counts, and whether a disclosed $2h_{\max}$ initialization target was covered; it still has no per-coordinate maturity mask. The option gate now requires 96 completed bars rather than the audited 60-bar baseline. A truncation audit over the final 32 bars confirms both the reason and the residual limitation: relative to full history, median/p90 IQR-normalized MAE is 0.501/2.009 at 60 bars, 0.035/0.931 at 96, 0.006/0.457 at 128, 0.0004/0.118 at 192, 0.00002/0.030 at 256, and approximately 0/0.003 at 365. Thus 96 is an explicit availability heuristic, not a convergence guarantee. The carrier baseline includes the current available observation. Sparse missing volume can leave participation windows unavailable; missing or zero volume makes the corresponding impact observation unavailable, although a rolling impact mean may still use earlier admissible observations. Equal visual row spacing differs from the log-horizon coordinate used in derivatives.

The v1 null state also requires careful language. If $P = v = G = 0$, then $Coh = 1$, state strength $S = 0$, disorder $E = 0$, Structure $= 0.42$, and the legacy organization channel $O = 0.68$. These values encode cross-horizon agreement and low motion, not active directional organization; the alternating-chop structure value 0.419 is essentially this floor. V1.1 exposes activity and horizon agreement separately, labels their composite, and reports the flat-field anchors without changing the v1 state vector. A gated alternative such as $S(0.58 + 0.42Coh)$ remains a candidate v2 ablation, not a silent reinterpretation of the reported v1 results.

The Form dictionary clusters aggregate state vectors rather than complete cloud shapes; identities are not stable across windows. Evaluation bars and future option windows are serially dependent and overlapping. The current outcome ledger does not adjust for duration, delta, moneyness, spread, commission, slippage, early exercise, or capacity. Scanner selection and discretionary closure can create selection and survivorship bias. Cached cross-market sources have heterogeneous vintages, and association is not causation. A current IV-minus-HV snapshot is not a historical volatility surface or arbitrage measure.

The next empirical stage should freeze an untouched symbol/period holdout, use rolling-origin evaluation with a purge and embargo at least as long as the outcome horizon (López de Prado, 2018), compare against price/volume, single-horizon EMA, HMM, change-point, wavelet/scattering, and persistent-homology baselines, and perform ablations by removing blending, higher derivatives, geometry/information, carriers, boundaries, and path fit. Option tests must normalize exposure and include executable quotes and costs. Dependence-aware intervals, an explicit trial registry, and a predeclared multiplicity target are prerequisites for any economic claim. Benjamini–Hochberg controls false discovery rate, not family-wise error rate; those targets must not be conflated.

7 Conclusion

Market Field is best understood as an auditable representation and research protocol. It converts completed OHLCV prefixes into a multiscale surface, derives bounded time/scale measurements, translates them through a restrained request-local dictionary, and records option-aligned context without granting it algorithmic ranking or execution authority. The present evidence exercises selected engineering properties and exposes calibration heterogeneity; visible advice can still influence human judgment and must be logged as exposure. The central hypothesis—that optionality mispricing behaves differently under different non-anticipative path configurations—remains deliberately unproven and is now instrumented for prospective evaluation.

Reproducibility Statement

The paper package contains the official unmodified ICLR 2026 style files, source main.tex, bibliography, generated tables and figures, input hashes, a cache-first asset script, executed notebooks, direct paper requirements, a full environment version capture, a clean-HEAD primary run receipt, and a supplementary receipt with per-file hashes for evaluated dirty-tree source. The script imports the same production functions used by the application. It records the paper as-of date and distinguishes provider retrieval from a same-machine cached rerun. Raw Yahoo Finance snapshots are not publicly redistributed; consequently a fresh clone can reproduce the code path and derived artifacts but cannot reconstruct the exact historical sample without an independently matching provider response. Commands and environment limitations are in the package README. No credentials, proprietary IBKR data, open positions, or secret-option records are included.

Ethics and Intended Use

This is an exploratory financial-research system, not investment advice or an autonomous trading agent. Visual fluency can create unwarranted confidence, so the interface and paper state which quantities are engineered analogues and which claims remain untested. The deployed option integration has no algorithmic ranking or execution authority, but its display may affect a human decision. Any later promotion requires new out-of-sample evidence, exposure-aware evaluation, and manual approval.

References

Ryan Prescott Adams and David J. C. MacKay. Bayesian online changepoint detection. arXiv preprint arXiv:0710.3742, 2007. URL https://arxiv.org/abs/0710.3742.

Yakov Amihud. Illiquidity and stock returns: Cross-section and time-series effects. Journal of Financial Markets, 5(1):31–56, 2002. doi: 10.1016/S1386-4181(01)00024-6. URL https://doi.org/10.1016/S1386-4181(01)00024-6.

Joakim Andén and Stéphane Mallat. Deep scattering spectrum. IEEE Transactions on Signal Processing, 62(16):4114–4128, 2014. doi: 10.1109/TSP.2014.2326991. URL https://doi.org/10.1109/TSP.2014.2326991.

Emmanuel Bacry, Jacques Delour, and Jean-François Muzy. Multifractal random walk. Physical Review E, 64(2):026103, 2001. doi: 10.1103/PhysRevE.64.026103. URL https://doi.org/10.1103/PhysRevE.64.026103.

Jushan Bai and Pierre Perron. Estimating and testing linear models with multiple structural changes. Econometrica, 66(1):47–78, 1998. doi: 10.2307/2998540. URL https://doi.org/10.2307/2998540.

David H. Bailey, Jonathan M. Borwein, Marcos López de Prado, and Qiji Jim Zhu. The probability of backtest overfitting. The Journal of Computational Finance, 20(4):39–69, 2017. doi: 10.21314/JCF.2016.322. URL https://doi.org/10.21314/JCF.2016.322.

Christoph Bandt and Bernd Pompe. Permutation entropy: A natural complexity measure for time series. Physical Review Letters, 88(17):174102, 2002. doi: 10.1103/PhysRevLett.88.174102. URL https://doi.org/10.1103/PhysRevLett.88.174102.

Jozef Baruník and Tomáš Křehlík. Measuring the frequency dynamics of financial connectedness and systemic risk. Journal of Financial Econometrics, 16(2):271–296, 2018. doi: 10.1093/jjfinec/nby001. URL https://doi.org/10.1093/jjfinec/nby001.

Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. Journal of the Royal Statistical Society: Series B (Methodological), 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x. URL https://doi.org/10.1111/j.2517-6161.1995.tb02031.x.

Tim Bollerslev, George Tauchen, and Hao Zhou. Expected stock returns and variance risk premia. The Review of Financial Studies, 22(11):4463–4492, 2009. doi: 10.1093/rfs/hhp008. URL https://doi.org/10.1093/rfs/hhp008.

Peter Carr and Liuren Wu. Variance risk premiums. The Review of Financial Studies, 22 (3):1311–1341, 2009. doi: 10.1093/rfs/hhn038. URL https://doi.org/10.1093/rfs/hhn038.

Ken Chung and Anthony Bellotti. Evidence and behaviour of support and resistance levels in financial time series. arXiv preprint arXiv:2101.07410, 2021. URL https://arxiv.org/abs/2101.07410.

Francis X. Diebold and Kamil Yilmaz. Better to give than to receive: Predictive directional measurement of volatility spillovers. International Journal of Forecasting, 28(1):57–66, 2012. doi: 10.1016/j.ijforecast.2011.02.006. URL https://doi.org/10.1016/j.ijforecast.2011.02.006.

Jean-Pierre Eckmann, S. Oliffson Kamphorst, and David Ruelle. Recurrence plots of dynamical systems. Europhysics Letters, 4(9):973–977, 1987. doi: 10.1209/0295-5075/4/9/004. URL https://doi.org/10.1209/0295-5075/4/9/004.

John F. Ehlers and Ric Way. Introducing SwamiCharts: A modern approach to technical indicator visualization. Technical Analysis of Stocks & Commodities, March 2012a. URL https://www.traders.com/Documentation/FEEDbk_docs/2012/03/Ehlers.html.

John F. Ehlers and Ric Way. Setting strategies with SwamiCharts: Examining the big picture. Technical Analysis of Stocks & Commodities, April 2012b. URL https://www.traders.com/Documentation/FEEDbk_docs/2012/04/Ehlers.html.

Marian Gidea and Yuri Katz. Topological data analysis of financial time series: Landscapes of crashes. Physica A: Statistical Mechanics and its Applications, 491:820–834, 2018. doi: 10.1016/j.physa.2017.09.028. URL https://doi.org/10.1016/j.physa.2017.09.028.

Teofilo F. Gonzalez. Clustering to minimize the maximum intercluster distance. Theoretical Computer Science, 38:293–306, 1985. doi: 10.1016/0304-3975(85)90224-5. URL https://doi.org/10.1016/0304-3975(85)90224-5.

Amit Goyal and Alessio Saretto. Cross-section of option returns and volatility. Journal of Financial Economics, 94(2):310–326, 2009. doi: 10.1016/j.jfineco.2009.01.001. URL https://doi.org/10.1016/j.jfineco.2009.01.001.

James D. Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. Econometrica, 57(2):357–384, 1989. doi: 10.2307/1912559. URL https://doi.org/10.2307/1912559.

Campbell R. Harvey, Yan Liu, and Heqing Zhu. … and the cross-section of expected returns. The Review of Financial Studies, 29(1):5–68, 2016. doi: 10.1093/rfs/hhv059. URL https://doi.org/10.1093/rfs/hhv059.

Lawrence Hubert and Phipps Arabie. Comparing partitions. Journal of Classification, 2(1): 193–218, 1985. doi: 10.1007/BF01908075. URL https://doi.org/10.1007/BF01908075.

Perry J. Kaufman. Smarter Trading: Improving Performance in Changing Markets. McGraw-Hill, 1995. ISBN 9780070340022.

Andrew W. Lo, Harry Mamaysky, and Jiang Wang. Foundations of technical analysis: Computational algorithms, statistical inference, and empirical implementation. The Journal of Finance, 55(4):1705–1765, 2000. doi: 10.1111/0022-1082.00265. URL https://doi.org/10.1111/0022-1082.00265.

Marcos López de Prado. Advances in Financial Machine Learning. Wiley, 2018. ISBN 9781119482086.

J. B. MacQueen. Some methods for classification and analysis of multivariate observations. In Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability, volume 1, pp. 281–297. University of California Press, 1967. URL https://digicoll.lib.berkeley.edu/record/113015.

Stéphane Mallat. Group invariant scattering. Communications on Pure and Applied Mathematics, 65(10):1331–1398, 2012. doi: 10.1002/cpa.21413. URL https://doi.org/10.1002/cpa.21413.

Norbert Marwan, M. Carmen Romano, Marco Thiel, and Jürgen Kurths. Recurrence plots for the analysis of complex systems. Physics Reports, 438(5–6):237–329, 2007. doi: 10.1016/j.physrep.2006.11.001. URL https://doi.org/10.1016/j.physrep.2006.11.001.

Jose A. Perea and John Harer. Sliding windows and persistence: An application of topological methods to signal analysis. Foundations of Computational Mathematics, 15(3):799–838, 2015. doi: 10.1007/s10208-014-9206-z. URL https://doi.org/10.1007/s10208-014-9206-z.

Dimitris N. Politis and Joseph P. Romano. The stationary bootstrap. Journal of the American Statistical Association, 89(428):1303–1313, 1994. doi: 10.1080/01621459.1994.10476870. URL https://doi.org/10.1080/01621459.1994.10476870.

Peter J. Rousseeuw. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. Journal of Computational and Applied Mathematics, 20:53–65, 1987. doi: 10.1016/0377-0427(87)90125-7. URL https://doi.org/10.1016/0377-0427(87)90125-7.

Charles Spearman. The proof and measurement of association between two things. The American Journal of Psychology, 15(1):72–101, 1904. doi: 10.2307/1412159. URL https://doi.org/10.2307/1412159.

Floris Takens. Detecting strange attractors in turbulence. In Dynamical Systems and Turbulence, Warwick 1980, volume 898 of Lecture Notes in Mathematics, pp. 366–381. Springer, 1981. doi: 10.1007/BFb0091924. URL https://doi.org/10.1007/BFb0091924.

Halbert White. A reality check for data snooping. Econometrica, 68(5):1097–1126, 2000. doi: 10.1111/1468-0262.00152. URL https://doi.org/10.1111/1468-0262.00152.

Jr. Wilder, J. Welles. New Concepts in Technical Trading Systems. Trend Research, 1978. ISBN 9780894590276.

A Complete Field Calculus

A.1 Input contract and pressure construction

The public research API accepts one symbol and one timeframe from {1m, 5m, 15m, 30m, 1h, 2h, 4h, 1D, 1W}. Horizons are bar counts within that timeframe; the current system does not fuse nine timeframes into one model. Requests permit 60–5,000 visible bars and horizons 4–160, subject to 120 horizon rows and 120,000 visible field cells. The API requests an additional hidden prefix of $\max(72, 2h_J)$ bars, computes on all returned bars, and then trims the response. This is a requested warm-up, not a guarantee that the provider supplies it.

Let $\epsilon = 10^{-9}$, let unqualified clip mean $\text{clip}_{[0,1]}$, and define $\phi(x) = x/(1+|x|)$. The unsigned numerical clip first maps NaN to 0, $+\infty$ to 1, and $-\infty$ to 0. For span q, every EWM in the field uses the pandas adjust=False recursion

$$\text{EWM}_q(x)_1 = x_1, \qquad \text{EWM}_q(x)_t = (1-\alpha_q)\,\text{EWM}_q(x)_{t-1} + \alpha_q x_t, \quad \alpha_q = \frac{2}{q+1}. \tag{5}$$

The normalizer coerces timestamps and OHLC values, requires finite strictly positive OHLC with $H_t \geq \max(O_t, C_t, L_t)$ and $L_t \leq \min(O_t, C_t, H_t)$, drops invalid price rows, keeps the last duplicate timestamp, sorts rows, and reports every rejection category. Boundary comparisons accept only a tight isclose tolerance (relative and absolute tolerance 10^{-12}) to avoid rejecting floating-point boundary noise. Invalid or missing volume does not discard a valid price bar; it becomes unavailable for volume-dependent carriers and its coverage is reported. At least 60 valid price rows must remain for the public field. This visible input-quality contract is part of semantic revision 1.1.

True range follows Wilder (Wilder, 1978):

$$TR_t = \begin{cases} H_t - L_t, & t = 1, \\ \max(H_t - L_t,\ |H_t - C_{t-1}|,\ |L_t - C_{t-1}|), & t > 1. \end{cases} \tag{6}$$

The true-range mean uses the available trailing $\min(h_j, t)$ observations. In Equation 1, $z_{j,t} = 0$ when that mean is zero. In Equation 2, $e_{j,t} = 0$ when the path sum is zero or M_{t-h_j} is unavailable. Otherwise the triangle inequality makes clipping mathematically redundant. Let $s_{j,t} = \text{EWM}_5(d_{j,t} e_{j,t})$. The first adjacent-row blend is

$$q_{j,t} = 0.68 s_{j,t} + 0.32 N_j(s)_t, \tag{7}$$

where $N_1(s) = s_2$, $N_J(s) = s_{J-1}$, and $N_j(s) = (s_{j-1} + s_{j+1})/2$ in the interior (with $N_1(s) = s_1$ when $J = 1$). After $x_{j,t} = \text{EWM}_3(q_j)_t$, the second blend is

$$P_{j,t} = 0.58 x_{j,t} + 0.42 \frac{x_{j-1,t} + 2x_{j,t} + x_{j+1,t}}{4}, \qquad 1 < j < J. \tag{8}$$

At an edge, the bracketed neighborhood becomes $(2x_{\text{edge}} + x_{\text{neighbor}})/3$; for $J = 1$, $P = x$. These convex operations preserve the pressure bound. Equal row adjacency, rather than distance in $\log h$, is the declared v1 smoothing model.

A.2 Base field channels

Write $u_j = \log h_j$ and let $\mathcal{G}_u$ denote the same nonuniform-grid derivative used by numpy.gradient(..., edge_order=1). Its endpoints are one-sided slopes. For an interior row, with $a = u_j - u_{j-1}$ and $b = u_{j+1} - u_j$,

$$(\mathcal{G}_u X)_j = -\frac{bX_{j-1}}{a(a+b)} + \frac{(b-a)X_j}{ab} + \frac{aX_{j+1}}{b(a+b)}. \tag{9}$$

Let $R_1 = \mathcal{G}_u P$, $R_2 = \mathcal{G}_u R_1$, $G = 1.35|R_1|/(1 + 1.35|R_1|)$, $L = 1.35|R_2|/(1 + 1.35|R_2|)$, and $T = \text{clip}(2.70|\Delta_t P|)$, where the first temporal difference is zero. Then

$$S = \text{clip}(1.8|P|), \qquad Q = \text{clip}(1.4|v|), \tag{10}$$

$$B = \text{clip}(0.42G + 0.33T + 0.25L), \qquad Coh = \text{clip}(1 - G/1.35), \tag{11}$$

$$E^{(0)} = \text{clip}(0.60(1 - Coh) + 0.40Q), \qquad E = \text{EWM}_4(E^{(0)}), \tag{12}$$

$$O = \text{clip}(0.48Coh + 0.32S + 0.20(1 - E)). \tag{13}$$

Here S, Q, B, Coh, E, O are state strength, velocity strength, boundary energy, coherence analogue, disorder analogue, and the legacy display channel serialized as confidence. The state-vector stratum called Structure is different from O. Under the zero field, $(S, Q, G, L, T, E) = (0, 0, 0, 0, 0, 0)$, $Coh = 1$, and $O = 0.68$; this means agreement at zero and low disorder, not active organization. Further display channels are

$$\text{Expansion} = \text{clip}(1.8\ \text{sign}(P)v), \tag{14}$$

$$\text{Contraction} = \text{clip}(-1.8\ \text{sign}(P)v), \tag{15}$$

$$\text{Reflectivity} = \frac{\log(1 + 4\,\text{clip}(0.50S + 0.22Q + 0.28B))}{\log 5}, \tag{16}$$

$$\text{Convection} = \text{clip}\left(B(0.45 + 0.55Q)\right). \tag{17}$$

These names are operational. Coherence is not spectral coherence; disorder is not generally information entropy; reflectivity is not optical reflectivity; and convection does not assert fluid dynamics.

A.3 Research strata and carriers

For any row field Y, define signed normalization and causal difference operators

$$\mathcal{N}_q(Y)_{j,t} = \phi\left(\frac{Y_{j,t}}{\mathrm{EWM}_q(|Y_j|)_t}\right), \qquad \mathcal{D}_q(Y)_{j,t} = \mathcal{N}_q(Y_{j,t} - Y_{j,t-1}), \tag{18}$$

with first difference and result zero when the denominator is at most ϵ. Thus $v = \mathcal{D}_{13}(P)$, $a = \mathcal{D}_{13}(v)$, $j = \mathcal{D}_{13}(a)$, and $s_4 = \mathcal{D}_{13}(j)$. The research scale channels are

$$g = \mathcal{N}_{13}(\mathcal{G}_u P), \qquad c = \mathcal{N}_{13}(\mathcal{G}_u \mathcal{G}_u P), \qquad m = \mathcal{D}_{13}(g), \tag{19}$$

where every EWM normalizes along time, row by row. Thus m is a bounded temporal difference of the bounded scale gradient, not a raw mixed partial.

Order-3 permutation entropy is evaluated separately on each pressure row. At time t, a stable sort maps $(P_{j,t-2}, P_{j,t-1}, P_{j,t})$ to one of $3! = 6$ possible ordinal patterns, with earlier positions winning ties. The plug-in distribution uses the most recent at most 24 observed pattern instances:

$$H^{\mathrm{perm}}_{j,t} = -\frac{\sum_{\pi \in S_3} p_{\pi,j,t} \log p_{\pi,j,t}}{\log 6}, \tag{20}$$

with $0 \log 0 = 0$. Before two pattern instances exist, the value is zero. This short-window plug-in estimator is descriptive and can be downward biased. A nine-timeframe sensitivity audit compares alternative instance windows with v1's 24: median entropy correlations are 0.451 (8), 0.630 (12), 0.606 (48), and 0.342 (96), with a minimum dataset correlation of -0.014. Window length is therefore a material model parameter rather than an innocuous variance setting. With bounded scale gradient g and velocity v,

$$c_{v,j,t} = \tanh\left(-\frac{v_{j,t} g_{j,t}}{g_{j,t}^2 + 0.08}\right), \tag{21}$$

$$Pr_{j,t} = \mathrm{clip}\left(1.8\sqrt{|v_{j,t}||g_{j,t}|}(0.45 + 0.55 Coh_{j,t})\right). \tag{22}$$

The constant 0.08 is an engineered regularizer. For an idealized translating front $P(u,t) = F(t - \beta u)$, $P_t = F'$ and $P_u = -\beta F'$, so $-P_t P_u = \beta (F')^2$. This supports the sign convention—positive indicates motion toward larger horizons—but amplitude deformation can produce the same analogue.

The row-level strata are

$$\begin{aligned}
\mathrm{Structure}_{j,t} &= \mathrm{clip}(0.58 S_{j,t} + 0.42 Coh_{j,t}), && (23)\\
\mathrm{Kinematics}_{j,t} &= \mathrm{clip}(0.18|v| + 0.24|a| + 0.27|j| + 0.31|s_4|)_{j,t}, && (24)\\
\mathrm{Geometry}_{j,t} &= \mathrm{clip}(0.32|g| + 0.28|c| + 0.20|m| + 0.20B)_{j,t}, && (25)\\
\mathrm{Information}_{j,t} &= \mathrm{clip}(0.72 H^{\mathrm{perm}}_{j,t} + 0.28 E_{j,t}). && (26)
\end{aligned}$$

Define horizon-weighted and arithmetic reductions

$$\mathcal{A}_h(Y)_t = \frac{\sum_{j=1}^J h_j Y_{j,t}}{\sum_{j=1}^J h_j}, \qquad \mathcal{A}_0(Y)_t = \frac{1}{J}\sum_{j=1}^J Y_{j,t}. \tag{27}$$

The five derivative coordinates use $\mathcal{A}_h$. Structure, Kinematics, Geometry, Information, and Propagation use $\mathcal{A}_0$, with

$$\mathrm{Propagation}_t = \mathcal{A}_0(Pr)_t, \qquad \mathrm{CascadeBias}_t = \begin{cases} \dfrac{\sum_j Pr_{j,t} c_{v,j,t}}{\sum_j Pr_{j,t}}, & \sum_j Pr_{j,t} > \epsilon, \\ 0, & \text{otherwise.} \end{cases} \tag{28}$$

Let $C_t^+ = \max(C_t, \epsilon)$ and $r_t = \Delta \log C_t^+$ with $r_1 = 0$. Realized variation is $RV_{j,t} = \sqrt{\sum_{i=t-h_j+1}^{t} r_i^2}$ over available observations, without annualization. The local scaling exponent is

$$\gamma_{j,t} = \text{clip}_{[-2,2]} \left[\mathcal{G}_u \log(RV + \epsilon)\right]_{j,t}, \qquad \text{ScalingExponent}_t = \text{clip}_{[-2,2]} \mathcal{A}_0(\gamma)_t. \tag{29}$$

It is not a Hurst exponent or multifractal spectrum. For equal per-bar squared variation, $RV_h \propto h^{1/2}$ and $\gamma \approx 0.5$ is the natural reference. A value near 0 means accumulated variation changes little across the local scale neighborhood (for example, one common shock dominates); 1 means approximately linear growth in RV_h; and a negative finite-difference estimate is a local stencil/startup irregularity rather than evidence of anti-persistence.

Let $\widetilde{V}_t = V_t$ when volume is finite and nonnegative, and let it be unavailable otherwise. Define the pointwise impact observation

$$Z_t = \begin{cases} |r_t|/(C_t^+ \widetilde{V}_t), & \widetilde{V}_t > 0, \\ \text{NaN}, & \text{otherwise}, \end{cases} \tag{30}$$

so missing, invalid, and zero volume make that impact observation unavailable rather than zero. The row inputs

$$U_{j,t} = \text{mean}_{i=t-h_j+1}^{t,\, \widetilde{V}_i \text{ available}} \widetilde{V}_i, \qquad I_{j,t} = \text{mean}_{i=t-h_j+1}^{t,\, Z_i \text{ available}} Z_i \tag{31}$$

use available observations only (pandas rolling means with min_periods=1); either mean is unavailable when its window has no admissible observation. Here I is Amihud-like (Amihud, 2002). For $X \in \{RV, U, I\}$ and $b = \max(34, 2h_J)$, the bounded carrier field and separately displayed direct ratio are

$$L_b(X)_{j,t} = \text{clip}\left[0.5 + 0.5 \tanh\left(2 \log \frac{X_{j,t} + \epsilon}{\text{EWM}_b(X_j)_t + \epsilon}\right)\right], \tag{32}$$

$$R_b(X)_{j,t} = \text{clip}_{[0,10]} \frac{X_{j,t} + \epsilon}{\text{EWM}_b(X_j)_t + \epsilon}. \tag{33}$$

The three state-vector carriers are $\mathcal{A}_0(L_b(RV))$, $\mathcal{A}_0(L_b(U))$, and $\mathcal{A}_0(L_b(I))$. The pandas EWM baseline is evaluated on the masked series and includes the current available observation. A direct ratio is unavailable wherever its current rolling input is unavailable; participation and impact ratios also require at least one positive-volume observation in the input. Missing internal carrier cells are filled with 0.5 only after this calculation to keep the optional retrospective clustering path finite: 0.5 is a neutral computational placeholder, not an observation, and must not be reported as a measured 1.0× ratio. Impact cannot be observed from a zero-volume bar.

B Dictionary Chronology and Semantics

Using Equation 27, the exact state vector is

$$\begin{aligned} \mathbf{x}_t = (&\mathcal{A}_h(P), \mathcal{A}_h(v), \mathcal{A}_h(a), \\ &\mathcal{A}_h(j), \mathcal{A}_h(s_4), \mathcal{A}_0(\text{Structure}), \mathcal{A}_0(\text{Kinematics}), \\ &\mathcal{A}_0(\text{Geometry}), \mathcal{A}_0(\text{Information}), \text{Propagation}, \text{CascadeBias}, \\ &\text{ScalingExponent}, \mathcal{A}_0(L_b(RV)), \mathcal{A}_0(L_b(U)), \mathcal{A}_0(L_b(I))) \in \mathbb{R}^{15}. \end{aligned} \tag{34}$$

The first five, next seven, and final three coordinates form derivative, field-transform, and OHLCV-carrier families.

For N available rows, v1 uses zero-based split indexes

$$t_E = \max(5, \min(\lfloor 0.60N \rfloor, N - 6)), \tag{35}$$

$$t_F = \min(w, \max(0, t_E - 40)), \qquad w = \max(34, 2h_J), \tag{36}$$

$$n_{cal} = \min\{\max(20, \lfloor (t_E - t_F)/3 \rfloor), \max(0, t_E - t_F - 20)\}, \tag{37}$$

$$t_C = t_E - n_{cal}. \tag{38}$$

Proper fit is $[t_F, t_C)$, held-out distance calibration is $[t_C, t_E)$, and evaluation is $[t_E, N)$. Thus the fit and calibration segments retain at least 20 rows when the input guards permit. Scaling, codebook selection, centroids, and transition grammar use proper fit only; state-conditional distance references use calibration; evaluation supplies displayed sequences and descriptive forward annotations.

For feature ℓ, proper-fit center is the median and initial scale is the interquartile range. If scale is at most ϵ, fallback order is 1.4826 times median absolute deviation, population standard deviation, and finally 1, each selected only when greater than ϵ. Let $z_{t\ell} = (x_{t\ell} - m_\ell)/s_\ell$. Squared metric distance is

$$d(\mathbf{x}_t, \boldsymbol{\mu})^2 = \frac{1}{15}\sum_{\ell \in \mathcal{D}}(z_{t\ell} - \mu_\ell)^2 + \frac{1}{21}\sum_{\ell \in \mathcal{S}}(z_{t\ell} - \mu_\ell)^2 + \frac{1}{9}\sum_{\ell \in \mathcal{C}}(z_{t\ell} - \mu_\ell)^2, \tag{39}$$

where family sizes are 5, 7, and 3. Equivalently, coordinates are multiplied by the square root of their displayed weight before Euclidean k-means and silhouette computation.

For each candidate $k = 2, \ldots, \min(5, \lfloor n_{fit}/n_{min} \rfloor)$, where $n_{min} = \max(20, \lceil 0.05 n_{fit} \rceil)$, the first seed is the fit point closest to the metric-space origin. Each later seed is the point farthest from its nearest existing seed; first row wins an exact tie. Lloyd updates run for at most 40 iterations, nearest-centroid ties choose the lowest index, and an empty cluster retains its previous centroid. Convergence requires unchanged assignments and centroids within absolute tolerance 10^{-12} and zero relative tolerance. A candidate is discarded if any cluster lacks n_{min} rows, any centroid separation is at most 10^{-6}, two centroids share the same coordinatewise 0.35-bin signature, or mean metric-space silhouette is below 0.25. The accepted candidate maximizes silhouette, with smaller k winning a tie; if none passes, $k = 1$. Final centroids are ordered lexicographically after rounding metric coordinates to 10 decimals. These choices are deterministic engineering rules, not evidence that the resulting Forms are natural latent states.

The reported legacy match transform uses the median of pooled calibration nearest-centroid distances (falling back to fit distances only if calibration is empty):

$$\text{match}_t = \exp\left(-d_t / \max(\text{median}(d^{\text{cal}}), \epsilon)\right). \tag{40}$$

It is a monotone resonance transform, not a probability; the state-conditional empirical tail rank in Equation 4 is the more transparent distance context. Evaluation outcomes use overlapping windows and are not independent. Form names summarize calibration-relative measurement profiles; they are not universal economic regimes or concepts learned independently of the feature design.

Asset	Bars	Forms	Silhouette	Tail n	Outside (%)
SPY	2,148	2	0.320	860	10.6
QQQ	2,148	1	0.000	860	4.4
IWM	2,148	1	0.000	860	6.6
TLT	2,148	1	0.000	860	3.3
GLD	2,148	1	0.000	860	6.6
USO	2,148	3	0.315	827	6.8
VNQ	2,148	1	0.000	860	1.5
BTC-USD	3,124	1	0.000	1,250	4.4
Pooled	18,160	–	–	7,237	5.5

Table 2: Daily-series dictionary diagnostics. Silhouette is zero when the support gate returns one Form. Tail n counts evaluation bars with at least 20 state-conditional calibration references. The table's "Outside" column preserves the v1 legacy label and means upper 5% calibration-distance tail.

C Completed-Bar Option Contract

The option snapshot retains at most 365 daily bars, requires at least 96, and treats a session as complete after 16:15 America/New_York. It uses horizons $\{12, 14, \ldots, 48\}$ and excludes

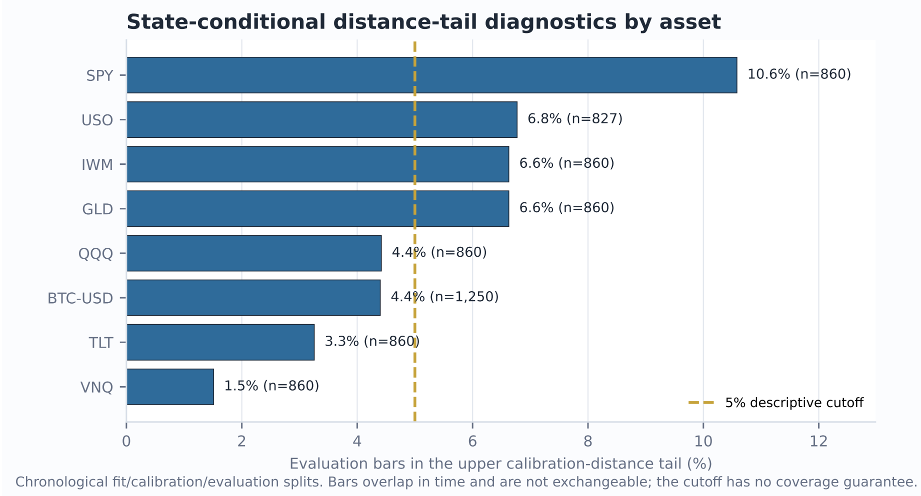

Figure 4: Asset heterogeneity in the empirical distance-tail diagnostic. The dashed line is a descriptive cutoff, not a promised coverage level.

retrospective dictionary sections. The 96-bar threshold is the disclosed $2h_{\max}$ initialization target adopted after the truncation audit; snapshots with 60–95 bars are unavailable and report how many bars are still needed. The payload also records alignment basis, maturity, input quality, and authority. It has no per-coordinate maturity mask, and 96 bars do not make the infinite-memory EWM identical to a full-history calculation. Support and resistance are strictly shifted:

$$\text{Support}_t = \min_{i=t-20}^{t-1} L_i, \qquad \text{Resistance}_t = \max_{i=t-20}^{t-1} H_i. \tag{41}$$

ATR is a trailing 14-bar simple mean of true range. Range position, close-to-SMA20, five-bar return, and ATR distance to the boundaries provide price-action context.

All hypothesis thresholds are computed from rows strictly before the current row and require at least 20 finite observations. Quantiles use NumPy's default linear interpolation:

- Organized expansion: structure $\geq q_{70}$, propagation $\geq q_{60}$, information $\leq q_{50}$, and $|p| \geq q_{60}$.
- Longward cascade: propagation $\geq q_{72}$, cascade bias $\geq \max(0.08, q_{62})$, and $|p| \geq q_{55}$.
- Geometry/disorder shock: geometry $\geq q_{75}$ and information $\geq q_{60}$.
- Kinematic exhaustion: kinematics $\geq q_{75}$, $|p| \geq q_{65}$, and pressure-aligned velocity $\widetilde{v} = \operatorname{sign}(p)v \leq q_{35}$.

These are deterministic exploratory definitions, not trained classifiers. The legacy fallback assumes a long directional single-leg interpretation; semantic v1.1 instead prefers signed delta or opening action plus option type, and abstains for unsupported exposure rather than applying that fallback to a known short or multi-leg structure.

Each scanner component is an existing heuristic on $[0, 100]$. Its base score remains

$$B = 0.30\,\text{cheapness} + 0.28\,\text{volEdge} + 0.27\,\text{contractQuality} + 0.15\,\text{executionQuality}. \tag{42}$$

For nonnegative integer recent-symbol, total-symbol, and recent-group counts (s_r, s_T, g_r), the pre-existing recurrence component is

$$\begin{aligned} R = \operatorname{clip}_{[0,100]}\big(&18\min(s_r, 4) + 4\min(\max(s_T - 1, 0), 5) \\ &+ 3\min(\max(g_r - s_r, 0), 6)\big), \qquad \text{rank} = \operatorname{clip}_{[0,100]}(B + 0.12R). \end{aligned} \tag{43}$$

Market Field has zero weight in both equations. In the manager, field evidence may lower displayed advisory confidence or increase displayed urgency; because the decision window is rebuilt after that advisory change, urgency may also recompute next_review_date. It cannot change hard vetoes, verdicts, targets, sizing, or execution. The challenger remains unpromoted until at least 100 cycles, 25 additional new cycles, incremental out-of-sample value, and manual approval. Existing targeted tests compare scanner scores and manager verdict/target-size fields with and without the field. The reported audit still lacks one complete eligibility-through-execution metamorphic proof and impression-level exposure logging.

D Cross-Market Shadow Atlas

The research page can read cached energy, real estate, agriculture, metals, sector-divergence, and crypto measures. For each source, it screens Spearman association (Spearman, 1904) between source pressure change and daily target log return at lags 0, 1, 5, and 20. The first 70% selects the largest absolute association. Holdout inference uses 199 fixed-block permutations of ranked source values with block length five and Benjamini–Hochberg correction (Benjamini & Hochberg, 1995). This controls false discovery rate under its assumptions, not family-wise error rate. A relationship is called persistent only when calibration and holdout signs agree, holdout $|\rho| \geq 0.15$, and $q \leq 0.10$.

This atlas is explicitly shadow_only with no field influence. It is association screening, not evidence of a spillover mechanism. Sources have heterogeneous cache vintages and are daily even when the selected field is intraday. The option snapshot intentionally excludes them until stronger alignment, provenance, and prospective validation exist.

E Multi-Timeframe and Resolution Sensitivity

The supplementary locally retained snapshot contains 9,235 completed bars across 15 datasets: SPY at 1m, 5m, 15m, 30m, 1h, 2h, 4h, 1D, and 1W, plus six additional daily markets. It excludes 15 latest rows under a conservative completion policy and finds zero duplicate timestamps, missing OHLCV cells, or invalid OHLC rows. All 46 prefix checks, including a direct mutation of the unseen SPY suffix, have zero error across 24,472 full-precision numeric values at tolerance 10^{-12}; all 16 repeated canonical-hash checks match. The audit covers live matrices and aggregate derivatives, strata, carriers, and ratios while bypassing response rounding. Every evaluated channel is finite across all nine timeframes. These are availability and implementation-invariance results, not evidence that one parameterization transfers economically across sampling frequencies.

For grid sensitivity, step 1 over horizons 8–64 is the reference and steps 2, 4, and 8 are recomputed from the same completed prefixes. The audit discards $W = \min(128, \max(60, \lfloor N/3 \rfloor))$ startup rows. For reference series y and candidate $\widehat{y}$, it reports

$$\mathrm{NMAE}_{IQR} = \frac{\mathrm{mean}_t \, |y_t - \widehat{y}_t|}{Q_{0.75}(y) - Q_{0.25}(y)}, \tag{44}$$

and leaves the statistic unavailable when the denominator is at most 10^{-9}. Across 15 datasets and 12 aggregate features, step 4 contains 180 pairs and produces median correlation 0.940 and median NMAE_{IQR} 0.203; its 95th-percentile error is 0.428. All three coarser grids together contain 540 pairs. The worst step-4 pair is the SPY 15m scaling exponent (NMAE_{IQR} 0.730, correlation 0.911). This does not define an optimal grid. It demonstrates that convergence must be measured per channel and that denser horizon resolution cannot be described as a purely visual enhancement.

Dictionary stability is weaker. On the 749-bar SPY daily window, the 70% prefix selects one Form while the full window selects two, producing adjusted Rand index (Hubert & Arabie, 1985) 0.000 on their overlap. The 85% prefix and full window both select two and achieve 0.873. On this shorter local window, SPY and QQQ upper calibration-distance-tail rates are 36.7% and 42.7%, versus 10.6% and 4.4% in the longer primary sample. The difference is not pooled away: it is direct evidence that Form identity and empirical distance ranks are window- and distribution-dependent. Only two next-state entries meet the current reliability support across seven daily dictionaries.

A live operational probe artifact records HTTP 200 for all nine timeframe selectors and its run-specific endpoint timings, but conservatively flagged all latest provider bars as possibly forming. Consequently, paper results use locally retained completed prefixes. Live endpoint availability and timing are reported only as an operational check, not a stable performance distribution.

F Evaluation Details and Threats to Validity

Synthetic construction. The deterministic reversal series contains 800 bars, a positive log drift before bar 400 and a negative drift afterward, with fixed OHLC ranges and volume. Threshold delay is the first post-reversal bar where $P_{h,t} < -0.05$. The trend/chop contrast holds bar ranges and deterministic generation fixed while changing directional path organization. These tests are useful precisely because the expected behavior follows from construction; they do not estimate generalization.

Prefix audit. For each of eight daily series and four cut points, the primary script rebuilds both the prefix and full history. It compares 11 four-decimal serialized channels over 19 horizon rows at the matching final prefix timestamp. The supplementary full-precision audit expands live numeric coverage as described above. Zero difference means the tested paths did not use the later suffix; it does not prove causal inference, provider timestamp correctness, or absence of bugs outside those paths. Remaining regression scope includes maturity metadata, hypothesis flags, lexicon fields under a fixed split, and the complete final option snapshot.

Distance tails. The pooled rate in Table 1 is descriptive. Bars are serially dependent; calibration sizes vary by state; the same hyperparameters were developed while observing related markets; and distributions drift. Asset-level dispersion in Figure 4 is more informative than the pooled proximity to 5%.

Operational benchmark. Timing uses one local Windows process with imports already loaded and cached in-memory histories. The supplementary distribution contains 63 warm runs across seven symbols; it excludes retrieval, persistence, fresh-process import time, concurrent load, network, and production queueing. It cannot be compared directly to production request latency or a service-level objective.

Data quality. Yahoo Finance adjusted OHLCV is convenient and reproducible only up to provider revisions. It is not the live IBKR path used by every application request. Corporate-action adjustments alter historical levels. The package therefore records canonical row hashes and never presents the sample as immutable exchange truth. Raw snapshots are retained locally but not redistributed; a fresh provider retrieval is a new sample even when its requested ticker and dates match.

G Research Progression and Design Rationale

1. Horizon strips. The starting point applied a common indicator across lookbacks, preserving SwamiCharts' visual scan across scale.
2. Continuous pressure. Indicator votes were replaced with a signed, efficiency-weighted displacement so direction and organization remained continuous.
3. Causal derivatives. Pressure change and successive bounded differences exposed strengthening, fading, and high-order instability without future bars.
4. Scale geometry and information. Derivatives over log horizon, boundary energy, and ordinal entropy separated across-scale fragmentation from within-scale motion.
5. Field cloud and scopes. A narrow horizon cloud became the overview; phase projections moved detailed relationships out of linear metric cards.
6. Translation layer. A deterministic request-local codebook attached compact names to measured profiles while exposing support and range evidence.

7. Context atlas. Prior-bar boundaries, option snapshots, and cached cross-market associations were added as visibly separate evidence layers.
8. Prospective options instrumentation. Immutable snapshots entered scanner and manager workflows under zero-weight safeguards, creating a path to future evaluation without silently changing decisions.

This progression is a design synthesis of established ideas, not a claim of new market physics. Its most important property is traceability: a phrase in the interface can be mapped back to a versioned vector, a completed data prefix, and explicit transformations.

H Planned Economic Evaluation

A preregistered next study should define a frozen universe, start date, option entry rule, holding horizon, and no-touch final holdout. At each origin, all scaling, thresholds, codebooks, and optionality baselines must be refit using only admissible history. The purge/embargo must cover the maximum holding horizon (López de Prado, 2018). Primary endpoints should be stated before inspection and include both statistical and economic measures. Dependence-aware resampling should use a documented block procedure such as the stationary bootstrap (Politis & Romano, 1994).

Baselines should include price/volume technical features, a single-horizon EMA system, HMM or sticky-HMM state estimates, structural or Bayesian change points, wavelet/scattering features, a sliding-window persistent-homology representation (Perea & Harer, 2015; Gidea & Katz, 2018), IV-minus-realized-volatility alone, and a cross-market connectedness model. Ablations should remove one family at a time and replace dictionary labels with the raw continuous vector.

Option outcomes require executable bid/ask observations, commissions, slippage, stale-quote controls, exercise conventions, and normalization by maturity, moneyness, delta, vega, exposure, and holding period. Dependence-aware bootstrap intervals and a predeclared multiplicity procedure must accompany repeated variants; the study must state whether it targets exploratory FDR or confirmatory family-wise error. A positive retrospective average is insufficient for promotion.

I Implementation Scope and Non-Claims

The current repository does not implement a Voss tactical wave, Takens embedding, optical flow, transfer entropy, Granger-causal propagation, Hurst or multifractal estimation, persistent cross-window Form identity, multi-symbol or nine-timeframe fusion, historical option surfaces, or a validated execution model. The renderer uses Canvas2D color quantization and display smoothing rather than true contour geometry, Gaussian/Sobel shaders, or WebGL.

Accordingly, the paper does not claim universal states, attractors, cycles, arbitrage, predictive return skill, profitable option selection, cross-market causation, or autonomous decision quality. The words causal and prospective refer respectively to non-anticipative prefix computation—not causal inference—and time-stamped future evaluation.

J LLM Usage Statement

OpenAI Codex materially assisted with repository inspection, formula extraction, literature organization, test and figure scripting, LaTeX drafting, and consistency checks. The author directed the research questions, system design, claim boundaries, and final editorial choices. Numerical results were generated by executable scripts against the repository implementation rather than produced by the language model. References were selected for primary-source relevance and should receive a final human bibliographic audit before formal submission. No language-model output is treated as empirical evidence.